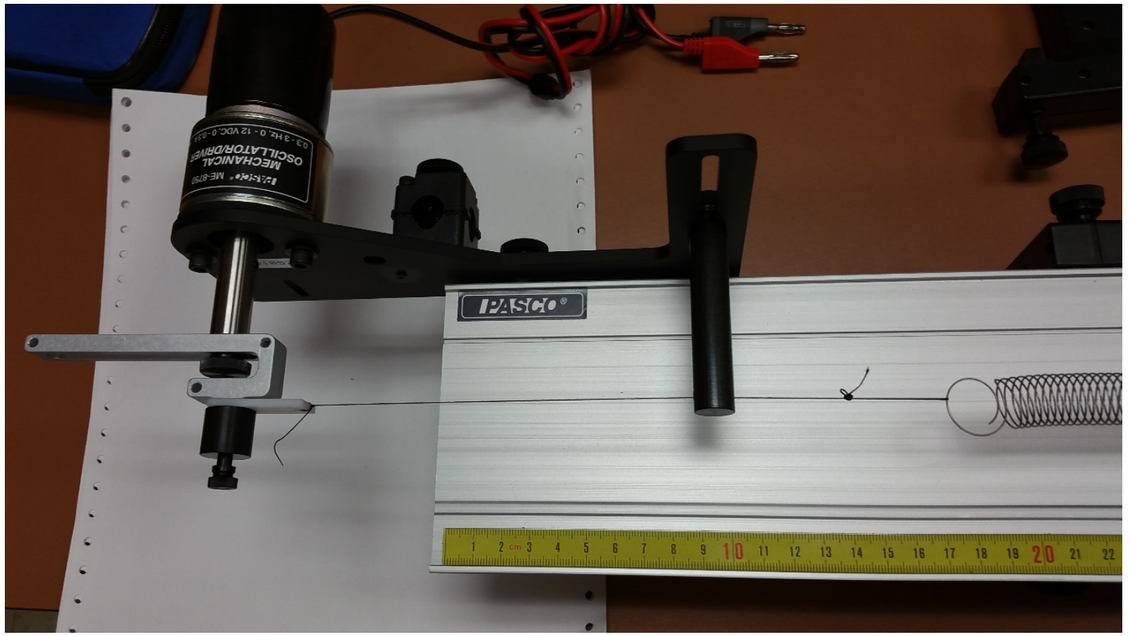
Forced vibration | Lab 5

Repeated motor motion can excite a structural resonance.

Measure impact decay, periodic response and displacement amplification.

MIE 402 · Prof. Ehsan Roohi · Fall 2026

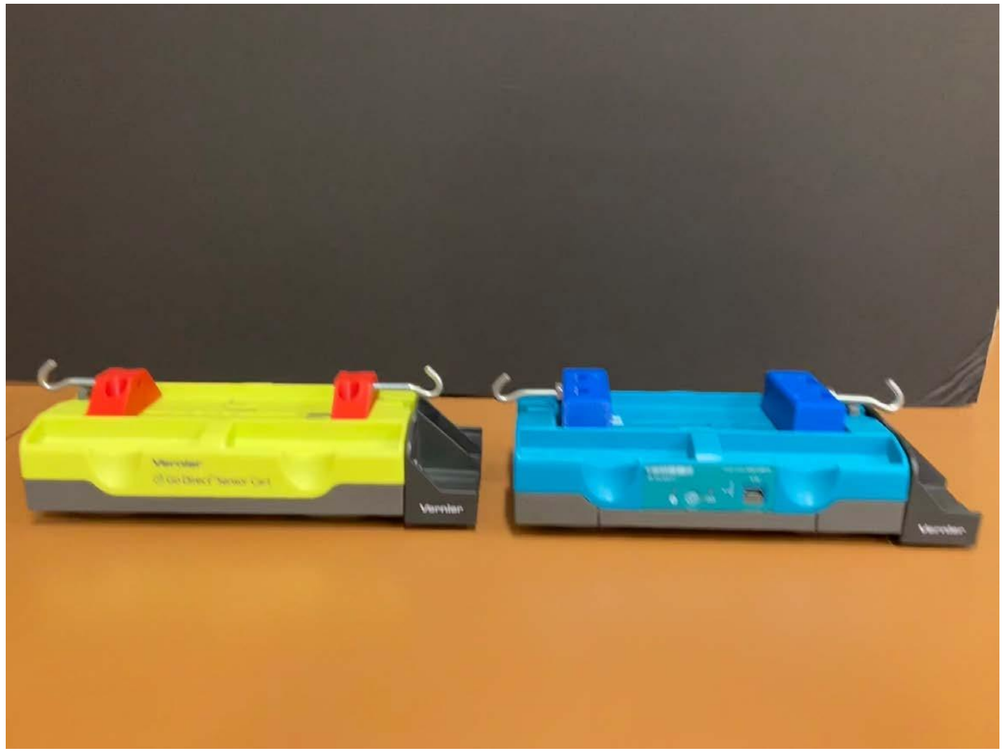


Engineering applications

Pump and motor supports: repeated loading can amplify vibration.

Precision stages: vibration limits positioning accuracy and settling.

This cart test identifies which drive speeds cause large motion.

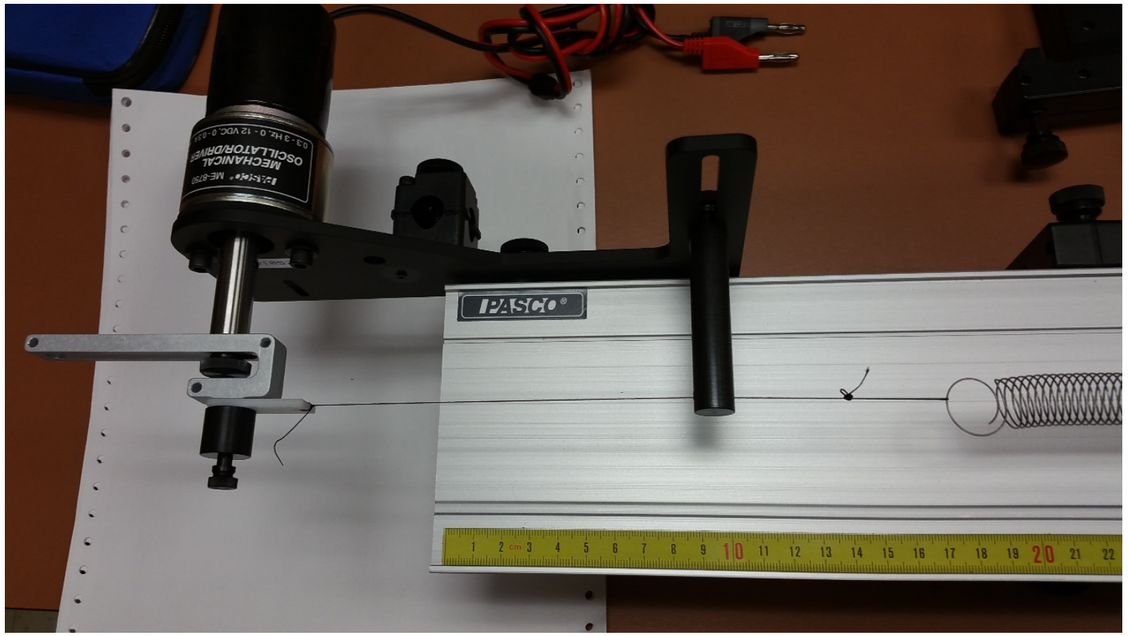


What the apparatus measures

The motor moves one spring end; the cart encoder measures cart position.

Two springs provide restoring force. Magnets provide eddy-current braking.

The tachometer measures drive speed; supply voltage only controls the motor.

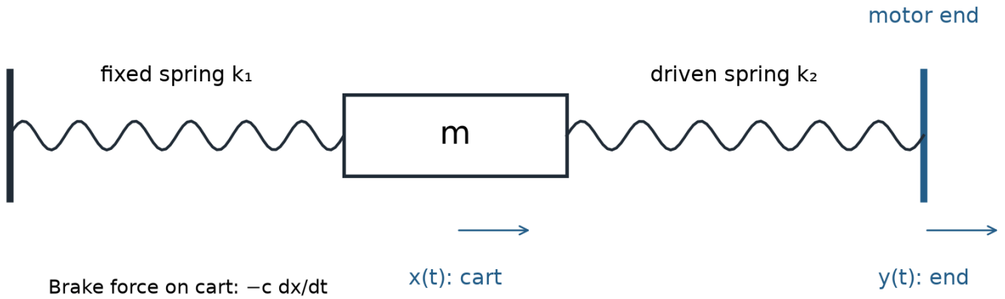


Coordinates and measured parameters

x(t): cart displacement; y(t): driven-end displacement, both positive right [m].

m: total moving mass [kg]; k_1, k_2: spring stiffness [N/m].

c: equivalent braking coefficient [N s/m], acting against the stationary track.

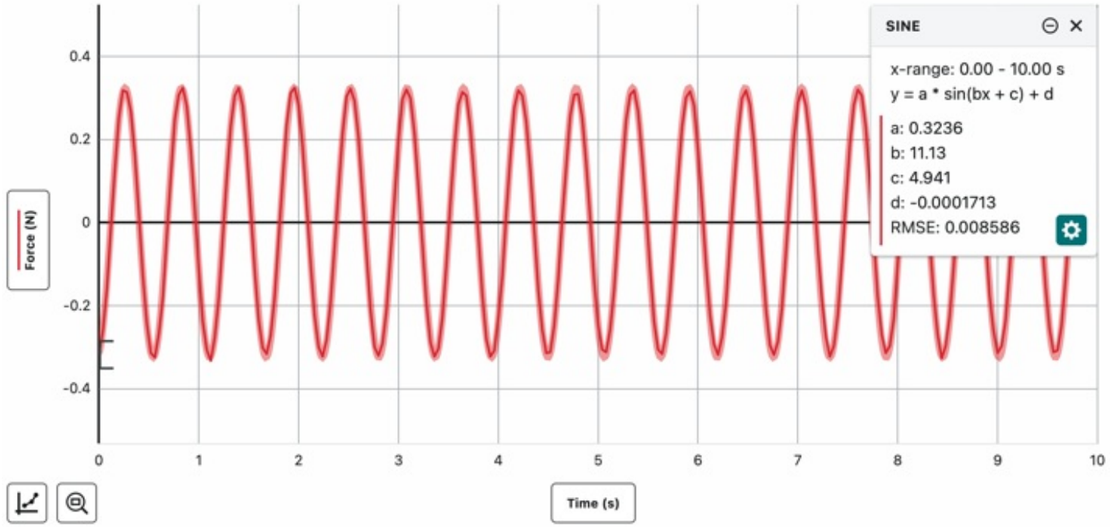


Spring and mass measurements

Weigh the cart, added hex rods and mounted dampers.

For a hanging calibration mass m_h, fit A sin(Bt + phase) + offset.

B is angular frequency [rad/s]; convert Hz before calculating stiffness.



$$k \approx m_h B^2, \qquad B = 2\pi f, \qquad k_{eq} = k_1 + k_2$$

Vernier Graphical Analysis

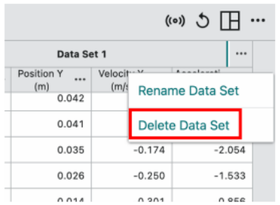
Connect the cart serial number and choose Position versus Time.

Set meters, collection rate at or below 25 Hz, and zero at equilibrium.

Collect before excitation. Export one continuous run as CSV; keep the raw file.







An impulse changes velocity

J: impulse [N s], the time-integral of the applied force F [N].

Velocities v^- and v^+ [m/s] are measured before and after the short impact; position is nearly continuous.

After contact ends, the system moves freely.

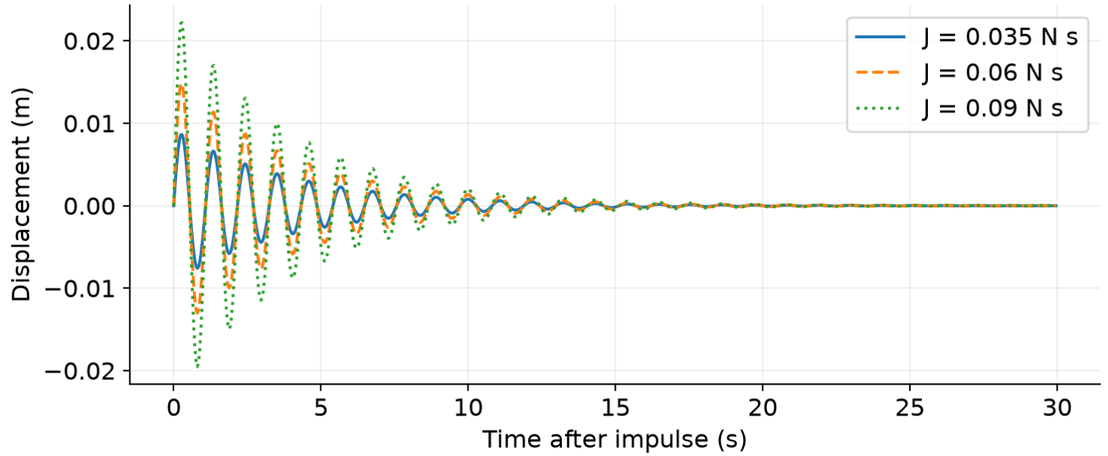
$$J = \int F\,dt, \qquad m(v^+ - v^-) = J$$

Impulse response follows the free-decay model

Define $\alpha = c/(2m)$ [1/s], $\omega_n = \sqrt{(k_e q/m)}$ [rad/s], $\omega d = \sqrt{(\omega_n^2 - \alpha^2)}$.

For a system initially at rest, the impulse sets initial speed J/m.

Three tap strengths test whether amplitude changes while frequency stays similar.



Illustrative model; not laboratory data

$$x(t) = \frac{J}{m\omega_d} e^{-\alpha t} \sin(\omega_d t)$$

What an impulse spectrum tells us

The time record reveals period and decay. The FFT is a frequency cross-check.

A finite decaying record spreads spectral energy; do not infer damping from peak height.

The fitted ωd and model ω_n are distinct; their difference is small for light damping.

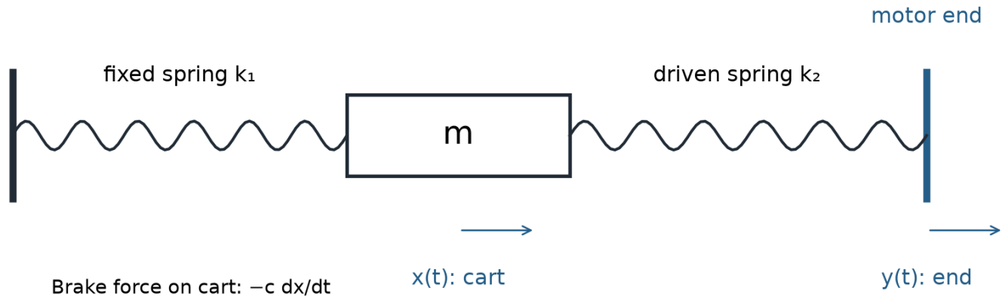
$$f_n = \frac{\omega_n}{2\pi}, \qquad f_d = \frac{\omega_d}{2\pi}$$

Forces from the moving spring end

x(t) is cart displacement. y(t) is the motor-end displacement. Both are horizontal coordinates, positive to the right [m].

The driven spring changes length by y − x, so its force on the cart is $k_2(y - x)$. The fixed spring contributes $-k_1x$.

Sum the two spring forces and the braking force, then collect the terms in x.



$$m\ddot{x} = -k_1x + k_2(y - x) - c\dot{x} \quad \Longrightarrow \quad m\ddot{x} + c\dot{x} + (k_1 + k_2)x = k_2y(t)$$

Why there is no derivative of y in this model

The magnetic brake reacts against the stationary track. Relative brake velocity is therefore the cart velocity.

Only the driven spring connects the cart to the moving endpoint. Its force depends on relative displacement.

A damper between the cart and moving endpoint would add a term proportional to the endpoint velocity. That is a different attachment.

$$F_{\text{brake}} = -c(\dot{x} - 0) \qquad \text{track brake;} \qquad F_{\text{moving damper}} = c(\dot{y} - \dot{x})$$

Prescribed input and settled output

The motor prescribes y(t) = Y cos(ωt). Y is the input peak amplitude [m]; ω = 2πf is drive angular frequency [rad/s].

A linear system in steady harmonic motion responds at the same drive frequency. Its amplitude and phase can differ.

X is cart peak amplitude [m]. ϕ is the output phase lag relative to input [rad]. We will solve for X and ϕ.

$$y(t) = Y\cos(\omega t), \qquad x_{ss}(t) = X\cos(\omega t - \phi)$$

Differentiate the assumed response

The first time derivative shifts cosine to negative sine and multiplies the amplitude by ω.

A second derivative returns cosine with a factor $-\omega^2$. Acceleration therefore opposes displacement.

Insert these derivatives into the force-balance equation from the preceding slides.

$$\dot{x}_{ss} = -\omega X \sin(\omega t - \phi), \qquad \ddot{x}_{ss} = -\omega^2 X \cos(\omega t - \phi)$$

Collect cosine and sine terms

Define $D = k_1 + k_2 - m\omega^2$ [N/m]. It represents the difference between stiffness and the inertial contribution.

Expand $\cos(\omega t - \phi)$ and $\sin(\omega t - \phi)$. Match coefficients of $\cos(\omega t)$ and $\sin(\omega t)$ on both sides.

The input has a cosine term only. The output coefficients must satisfy both equations below.

$$DX\cos\phi + c\omega X\sin\phi = k_2 Y, \qquad DX\sin\phi - c\omega X\cos\phi = 0$$

Solve for the amplitude ratio

Square the two coefficient equations and add them. The cross terms cancel and $\cos^2\phi + \sin^2\phi = 1$.

This gives $X^2[D^2 + (c\omega)^2] = (k_2Y)^2$. Take the positive square root because X and Y are amplitudes.

The input amplitude still determines X. Dividing by Y creates a ratio, so Y cancels only in X/Y.

$$\frac{X}{Y} = \frac{k_2}{\sqrt{(k_1 + k_2 - m\omega^2)^2 + (c\omega)^2}}, \qquad X = Y\left(\frac{X}{Y}\right)$$

Phase lag and the analytical time response

The two coefficient equations also determine the phase. Use atan2 to select the correct quadrant.

Below resonance the lag is usually small. At D = 0 it is π/2. At high frequency it approaches π.

The analytical waveform needs a known input phase as well as the amplitude. An output-only record does not measure the input/output lag.

$$\phi = \operatorname{atan2}(c\omega, k_1 + k_2 - m\omega^2), \qquad x_{ss}(t) = X\cos(\omega t - \phi)$$

Complex amplitudes give the same result

Define $i^2 = -1$. A sinusoid can be written as the real part of a complex amplitude times $\exp(i\omega t)$.

Differentiation then multiplies the complex amplitude by $i\omega$, and a second derivative multiplies it by $-\omega^2$.

The complex ratio H contains both gain and phase: its magnitude is X/Y and its argument is the negative of the lag.

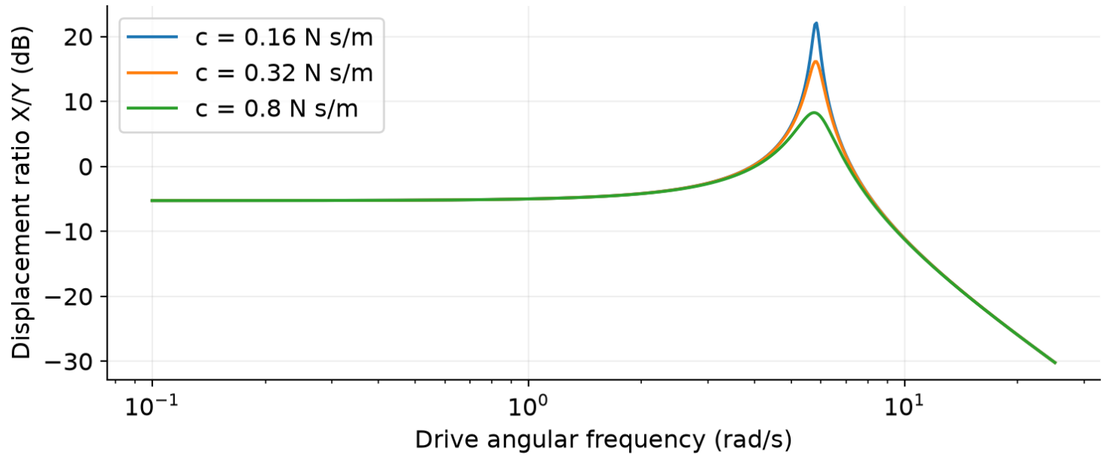
$$H(i\omega) = \frac{\hat{x}}{\hat{y}} = \frac{k_2}{k_1 + k_2 - m\omega^2 + ic\omega}$$

Low-frequency and resonance limits

At low drive frequency, the gain approaches $k_2/(k_1 + k_2)$. Both springs share the endpoint displacement.

Near resonance, inertia partly cancels restoring stiffness. Damping limits the amplitude. The maximum shifts below the natural frequency when damping is nonzero.

ζ is the dimensionless damping ratio. At high drive frequency the cart inertia reduces the displacement ratio.



Illustrative model; not laboratory data

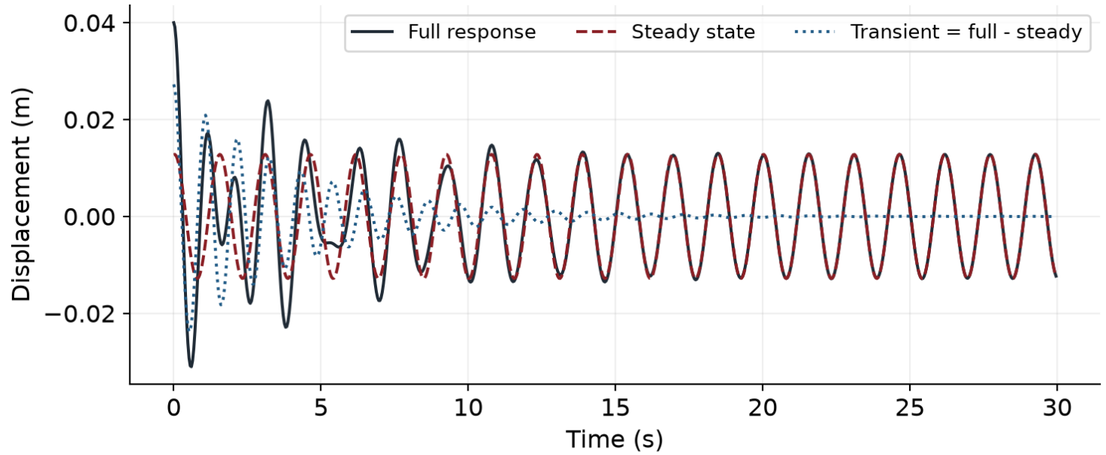
$$\omega_n = \sqrt{\frac{k_1 + k_2}{m}}, \qquad \zeta = \frac{c}{2\sqrt{m(k_1 + k_2)}}$$

The full response includes startup motion

The steady response solves the forced equation; a transient solves the same equation with zero right-hand side.

Their sum satisfies the full equation. The initial position and velocity determine the transient coefficients.

With positive damping, the transient decays and only the forced sinusoid remains. Select that settled interval for the gain comparison.



Illustrative model; not laboratory data

$$x(t) = x_{tr}(t) + x_{ss}(t)$$

How initial conditions determine the transient

Define $\alpha = c/(2m)$ [1/s] and the underdamped frequency $\omega d = \sqrt{[(k_1+k_2)/m - \alpha^2]}$ [rad/s].

A and B are transient displacement coefficients [m]. Set them so the total response matches the initial position and velocity.

Changing release conditions changes A and B. It does not change the final harmonic gain for fixed system parameters and drive.

$$x(t) = e^{-\alpha t}[A\cos(\omega_d t) + B\sin(\omega_d t)] + X\cos(\omega t - \phi)$$

Transfer function and state equations agree

A force-input transfer function gives displacement per force [m/N]. Here the effective input force is $k_2 y$.

Multiplying the force-input transfer function by k_2 gives the displacement ratio H [dimensionless].

Set velocity v = dx/dt [m/s]. Dividing Newton's law by m gives the second state equation below.

$$\dot{x} = v, \qquad \dot{v} = -\frac{k_1 + k_2}{m} x - \frac{c}{m} v + \frac{k_2}{m} y$$

Measure input travel and motor speed

Measure actual endpoint peak-to-peak travel, typically 2–3 cm, then divide by 2.

Use the tachometer for each run. One revolution corresponds to one drive cycle.

Record the drive amplitude again when the pin setting or endpoint travel changes.

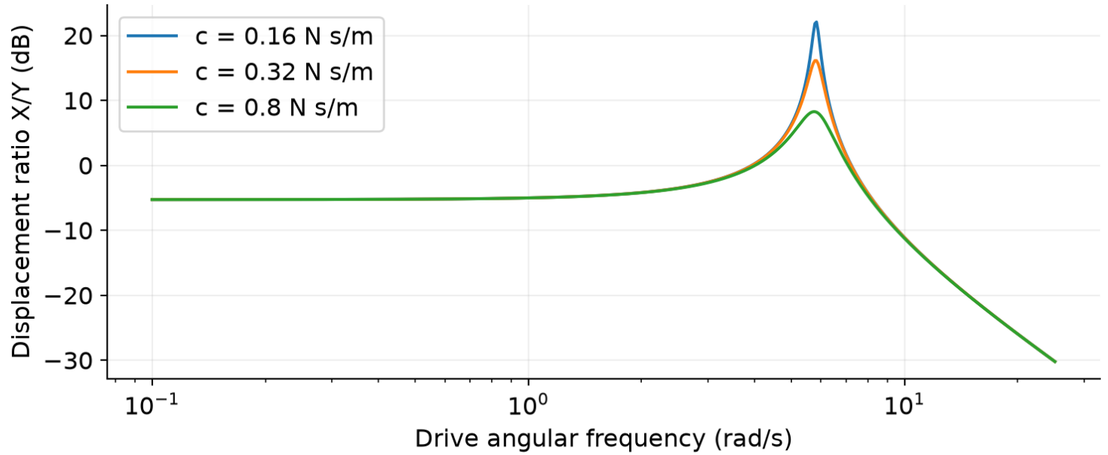
$$Y = \frac{y_{pp}}{2}, \qquad f = \frac{\text{rpm}}{60}, \qquad \omega = 2\pi f$$

A useful sweep covers both sides of resonance

Use 3–5 safe frequencies below and 3–5 above the expected resonance.

Allow startup to decay; record at least five settled cycles, preferably ten.

Keep the springs taut and travel safe. Stop if oscillations become excessive.



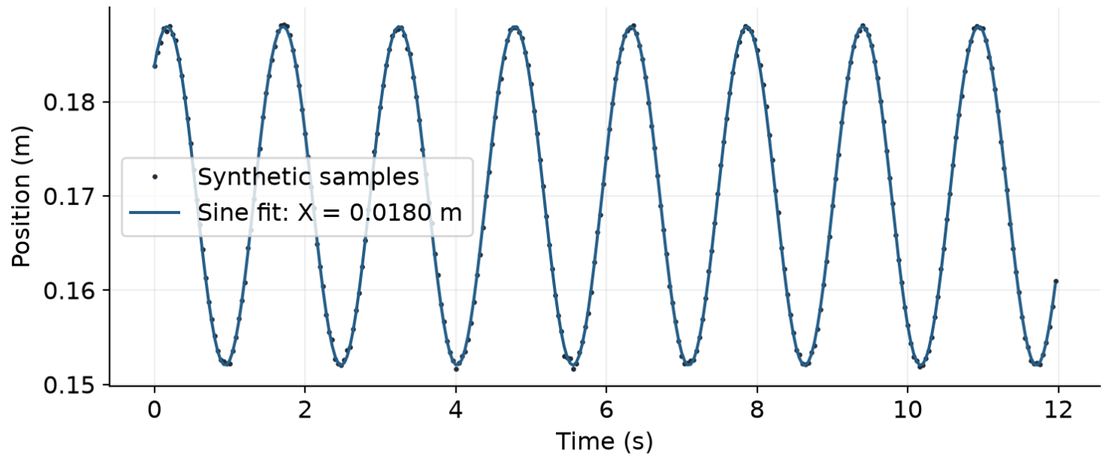
Illustrative prediction used to plan test points

Fit amplitude at the measured drive frequency

Fit $x(t) = \text{offset} + \text{slow drift} + a\cos(\omega t) + b\sin(\omega t)$.

The frequency comes from the tachometer; the code solves for a and b by linear least squares.

The peak amplitude is $X = \sqrt{(a^2 + b^2)}$. Inspect the fit against the data.



Synthetic example of measured-point and fit display

$$G = \frac{X}{Y}, \qquad G_{dB} = 20\log_{10} G$$

Choose damping for the tested configuration

Magnetic dampers change both damping and moving mass.

With the drive off and both ends stationary, record a free decay in the forced-test configuration.

Analyze that decay for c. Use it with the matching mass in the harmonic prediction.

$$c = 2m\alpha, \qquad \omega_d = \sqrt{\frac{k_1 + k_2}{m} - \alpha^2}$$

MATLAB workflow

Open Run_Lab5_Forced_Response.m. Enter measured mass, both stiffnesses and matching damping.

Select CSV/Excel files, a settled time interval and position units. Enter measured motor rpm and input peak-to-peak travel.

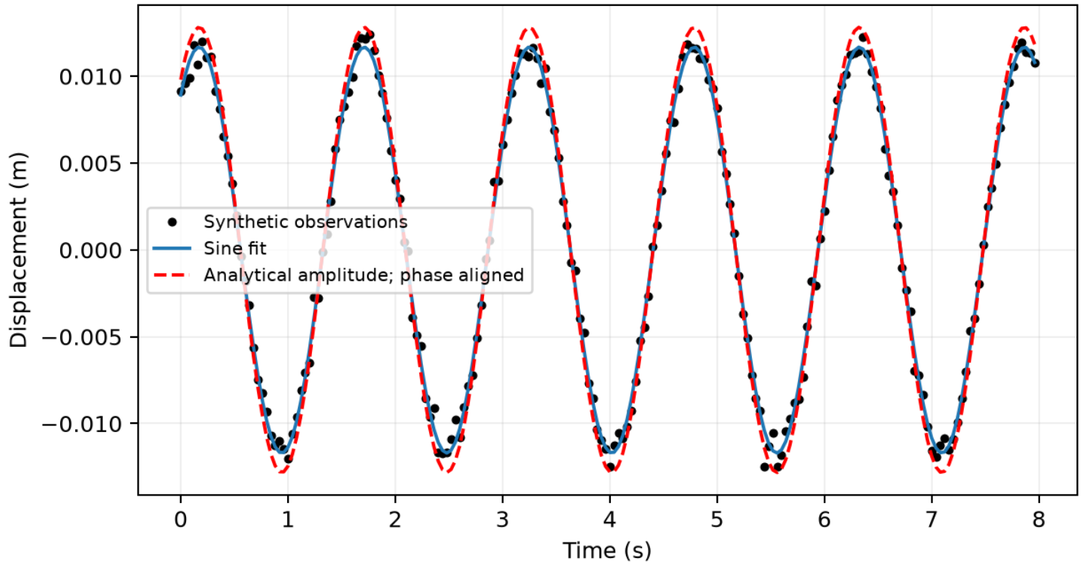
Each run exports measured data, its sine fit and the analytical amplitude comparison. The sweep figure compares measured gain with theory.

What the code compares with theory

The blue sine fit estimates the measured amplitude X. The red dashed curve uses amplitude predicted from m, k_1, k_2, c, Y and drive frequency.

Its phase is aligned to the output for display. That comparison tests amplitude; it does not measure input/output phase.

Read measured X, predicted X and percent difference in the summary. Compare X/Y across drive frequencies.



Synthetic example: fit and analytical amplitude are different quantities

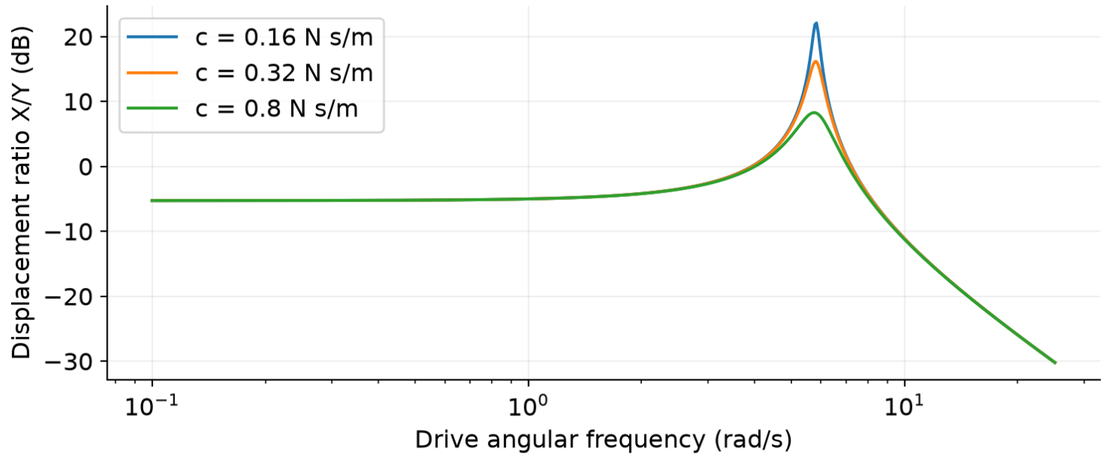
$$X_{\text{theory}} = \frac{k_2 Y}{\sqrt{(k_1 + k_2 - m\omega^2)^2 + (c\omega)^2}}$$

Figures for the combined Labs 4 + 5 report

Three impulse time histories and spectra: compare frequency and amplitude.

Representative harmonic record: identify startup and the settled interval.

Measured displacement ratios overlaid on theory using your measured parameters.



Illustrative theory only; add your own measured points

Check before leaving the laboratory

Show one impulse trace and spectrum, one settled sine fit and at least five sweep points to the GTA.

Confirm time units, meters, kilograms, peak versus peak-to-peak, and rpm conversion.

Copy the raw files and parameter log to every group member.

Three frequencies to distinguish

Natural frequency: set by mass and the two spring stiffnesses.

Damped transient frequency: appears in decay after release or impact.

Drive frequency: remains in the settled forced oscillation.

$$\omega_n \neq \omega_d \neq \omega \quad \text{in general}$$